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DYNAMIC CONTEXT- AWARE TRIES FOR KNOWLEDGE GRAPH CONSTRAINED LLM GENERATION

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Introduction

Large Language Models (LLMs) generate fluent but sometimes factually incorrect outputs (“hallucinations”) because they are probabilistic next-token predictors rather than grounded knowledge systems, limiting their reliability in knowledge-intensive applications (Huang et al., 2023; Ji et al., 2023).



Problem Statement

- Mitigation methods (prompting, RAG, agents): Limited guarantees or high computational cost (Lewis et al., 2020; Jiang et al., 2023)
- Graph-Constrained Reasoning (GCR): Restricts decoding to valid KG paths, reduces hallucination (Luo et al., 2024)
- Limitation: assumes static KGs
Not suited for dynamic, frequently updated graphs
Need adaptive, context-aware constraints at inference time (Sun et al., 2024; Luo et al., 2024)



Project Objectives

The objectives of this project are to:

- **Design** a Dynamic Context-Aware Trie (DCAT) mechanism for KG-constrained LLM decoding.
- **Implement** DCAT using token-level constrained decoding in an open-source LLM framework.
- **Evaluate** DCAT against unconstrained decoding (and a small-graph static baseline) in terms of validity, efficiency, and scalability.
- **Analyze** the trade-offs between dynamic constraint reconstruction and static pre-compiled constraints.



Expected Outcome(s)

At the end of the project, the following outcomes are expected:

- A working DCAT constrained decoding module for LLMs operating over KGs.
- Quantitative evidence showing reduction in invalid KG actions during generation.
- Performance analysis of runtime and memory costs under varying KG sizes and branching factors.
- A documented experimental framework and final project report.



METHODS TO BE USED

This study adopts an **experimental systems research approach** consisting of:

- constrained decoding design
- KG-based task formulation
- empirical evaluation and comparison with baselines.
- The project focuses on **decoding-time constraint enforcement**.

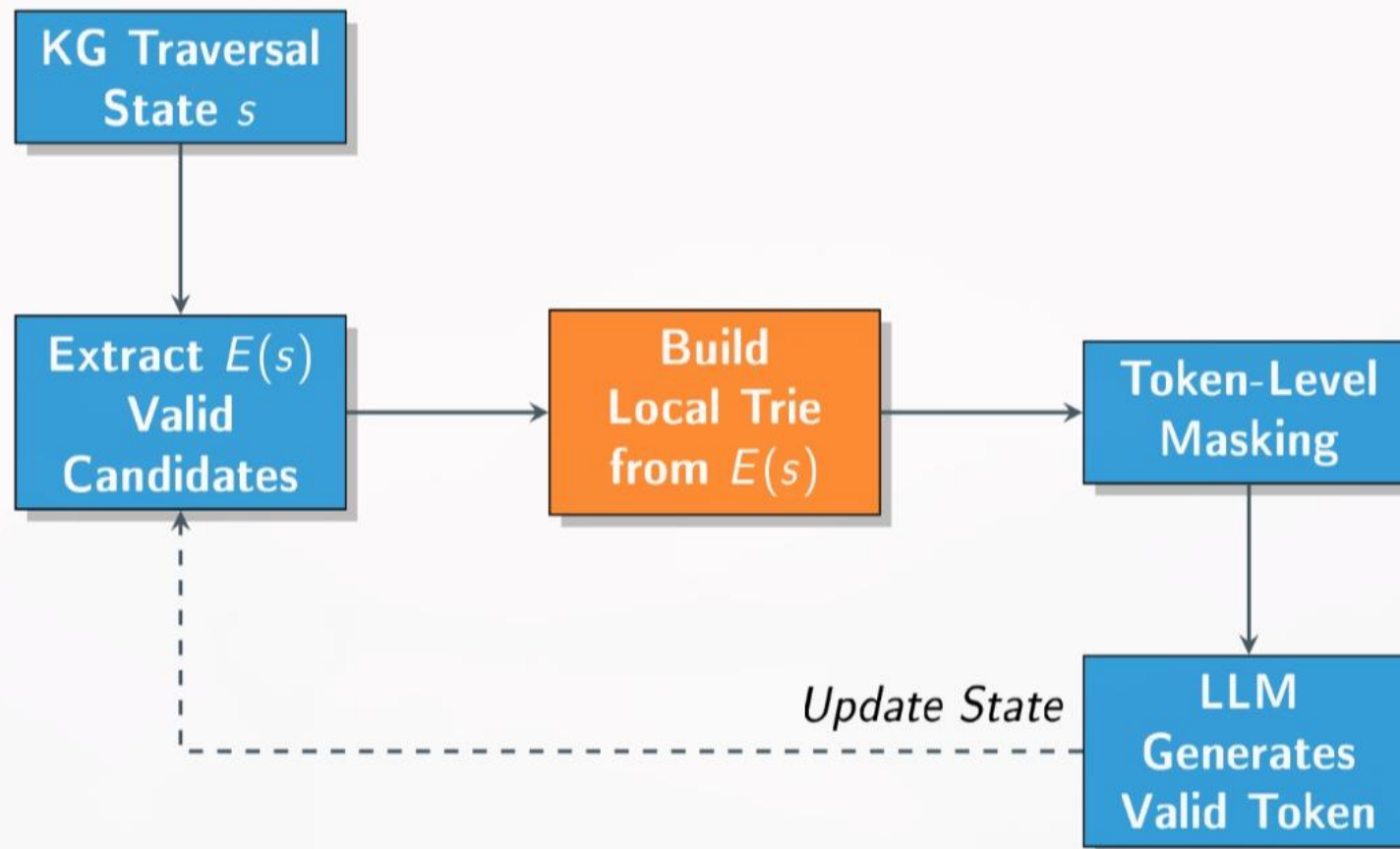


METHODS AND PROCEDURES

- Represent a Knowledge Graph using adjacency structures.
- At each traversal step:
 - retrieve valid next entities from the KG,
 - tokenize candidate entities using the LLM tokenizer,
 - build a **local trie** over token sequences.
- During decoding:
 - mask invalid tokens using constrained decoding hooks,
 - allow generation only along trie-valid prefixes.
- Update traversal state and rebuild constraints dynamically.
- Log validity, latency, and memory usage for evaluation.



METHODS AND PROCEDURES





Facilities and Tools to be Used

Hardware / Environment

- Personal computer with GPU

Software Tools

- Python
- Hugging Face transformers
- PyTorch
- NetworkX / custom KG data structures
- Jupyter Notebook for experiments

Datasets

- FB15k-237 or a Wikidata-derived KG subset



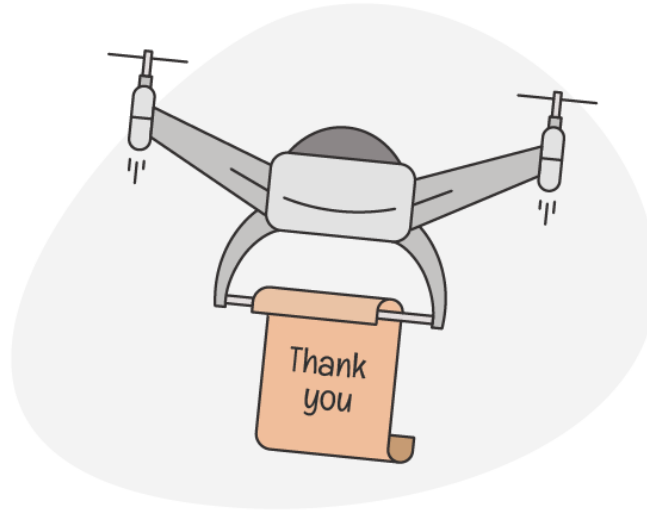
Project Protocol

Project activities	Semester I	Semester II	Remarks
Proposal Write-up	04/02/26 - 06/02/26		Completed
Proposal Defense	12/02/26		Current Stage
Writing :			
Chapter 1	16/02/26 - 22/02/26		Pending
Chapter 2	23/02/26 - 09/03/26		Pending
Chapter 3	12/03/26 – 26/03/26		Pending
Chapter 4		09/06/26 -26/06/26	Pending
Chapter 5		28/06/26 -13/07/26	Pending
Submission of project Work Reports	31/03/26	Pending	Pending



References

- Hogan et al.,(2021) "*Knowledge Graphs*", ACM Computing Surveys ,Vol.54, No.4, 71pp.
- Pan et al.,(2024)," *Unifying LLMs and Knowledge Graphs*", IEEE,Vol.36 ,pp.1-5.
- Huang et al.,(2025) "*Survey on Hallucination in LLMs*", ACM TOIS, Vol.43, No.2 ,42pp.
- Willard and Louf(2023) "*Efficient Guided Generation for LLMs*" , arXiv, Vol.4 ,18pp.
- Luo et al., (2024) "*Graph-Constrained Reasoning*", arXiv, Vol.2, 26pp.
- Li et al., (2024) "*Decoding on Graphs*", arXiv, Vol. 1, 15pp.



Comments and Questions Please

